Ultimate Solution to Why an Apple Falls

Derivation of the Essence of Gravity

From Apple Falling to the Cosmic Refresh Engine

Abstract

Within the framework of TCM-NAN(λ) (Time Creation Model – Nothing Allows Nothing), this paper systematically reconstructs the essence of gravity. We abandon the pictures of "action at a distance" and "spacetime curvature" in traditional physics, and redefine gravity as the projection of the gradient of the refresh delay field λ on macroscopic scales. By deriving the microscopic mechanism of the classic phenomenon of "an apple falling", we prove that gravitational acceleration arises from the "refresh pressure difference" experienced by internal refresh units of an object in the gradient of the λ field. The paper further clarifies that from the weak-field limit of Newton's law to the strong-field effects of general relativity, and even to galactic rotation curves, all can be unified under λ and its differential forms.

One of the oldest questions in physics: *Why does an apple fall?* Newton told us that it is because universal gravitation exists between the Earth and the apple, whose magnitude is proportional to the product of masses and inversely proportional to the square of distance. Einstein further argued that it is not a force at all, but that massive objects curve surrounding spacetime, and the apple moves along the "shortest path" (geodesic) in curved spacetime.

However, neither explanation answers a more fundamental question: What is the propagation medium for the "curvature" of spacetime or the "force" of gravity? Without a medium, how can two objects "sense" each other's existence? In the new world of TCM-NAN(λ), we provide a completely new answer.

Part I: Foundation of the New World – The Refresh Delay Field λ

In TCM-NAN(λ), the universe is not a curved manifold, but an absolutely stationary, discrete, rigid coordinate grid we call the "whiteboard". Time is not a dimension, but a state-update sequence of local grid points. Each grid point "refreshes" at a base frequency (a cycle from nothing to existence and back to nothing).

1.1 Definition of the Refresh Delay Field λ

The presence of mass locally slows down this refresh process. An object of mass M induces a refresh delay field $\lambda(r)$ at distance r from its center of mass. It is not a particle, but a scalar field representing the extra time that a single refresh takes compared to the vacuum baseline.

Its core formula:

$$\lambda(r) = T_0 \cdot \frac{GM}{c^2 r} \tag{1}$$

Where:

- T_0 = vacuum baseline refresh period (on Planck scale $\approx 10^{-44}$ s)
- G = Newtonian gravitational constant
- c = speed of light
- r = distance to the center of mass

Physical meaning: $\lambda(r)$ is a preprinted "refresh step" gradient on the whiteboard. Closer to the mass center (smaller r) means larger λ, slower refresh, and slower time flow; farther away means faster refresh.

Part II: Microscopic Mechanism of Apple Falling – "Refresh Pressure Difference"

We now consider an apple (composed of many microscopic refresh units) placed in the λ field produced by the Earth. The apple does not fall because it is "pulled" toward the Earth, nor does it slide along "curved" spacetime – it is **pushed** toward the Earth by a pressure arising from differences in refresh gradient.

2.1 Internal-External Refresh Conflict: From Velocity to Position

The apple is a high-λ region (slow refresh). The surrounding vacuum background is a low-λ region (fast refresh). This creates an "internal-external refresh conflict": fast outside, slow inside. How does this "time mismatch" appear spatially?

In TCM, particles (such as electrons in the apple) undergo a high-frequency "disappear-reappear" cycle. In each reappearance, the system must determine its new position on the whiteboard. This position is jointly determined by the wavefunction phase at the previous step and the local refresh delay λ.

1. Phase lag accumulation: Consider one constituent unit of the apple. The side near the Earth (distance r) experiences larger $\lambda(r)$ (slower refresh), while the side far from the Earth (distance $r+d$) experiences smaller $\lambda(r+d)$ (faster refresh). A tiny difference in single-refresh time exists between the two sides:

$$\delta t_{\text{diff}} = \lambda(r) - \lambda(r+d) \approx -\frac{d\lambda}{dr} \cdot d \tag{2}$$

2. Position drift: To preserve wavefront continuity (a fundamental requirement of the refresh algorithm), this time difference δt_{diff} converts into a spatial lateral drift Δx per refresh cycle:

$$\Delta x \approx c \cdot \delta t_{\text{diff}} \tag{3}$$

3. Macroscopic acceleration: This microscopic, Earth-directed drift Δx occurs $f_0 = 1/T_0$ times per second, and accumulates macroscopically into the measured acceleration g .

2.2 Macroscopic Formula of Path Deflection: Necessity of Fermat' s Principle

The mechanism above can be expressed more concisely at the macroscopic level. In an inhomogeneous refresh medium, motion follows the principle of minimal refresh resistance, and acceleration is determined by the gradient of the λ field:

$$\boldsymbol{a} = -c^2 \nabla(\ln \lambda) \tag{4}$$

This formula is the core of the entire TCM gravity theory.

Example: Mathematical Derivation of Apple Falling

- Step 1: Construct the λ field of the Earth.

Substitute Earth mass M into $\lambda(r) = 1 + \frac{GM}{c^2 r}$ (vacuum baseline $\lambda = 1$ explicitly written).

- Step 2: Compute the gradient of $\ln \lambda$.

Under weak-field conditions at Earth' s surface, $\frac{GM}{c^2 r} \approx 10^{-9} \ll 1$. Taylor expansion:

$$\ln \lambda = \ln\left(1 + \frac{GM}{c^2 r}\right) \approx \frac{GM}{c^2 r} \tag{5}$$

- Step 3: Substitute into acceleration formula.

$$\boldsymbol{a} = -c^2 \nabla\left(\frac{GM}{c^2 r}\right) = -\nabla\left(\frac{GM}{r}\right) = -\frac{GM}{r^2}\hat{\boldsymbol{r}} \tag{6}$$

- Result: We exactly recover Newton’s law of universal gravitation. For Earth ($M \approx 5.97 \times 10^{24}\ \mathrm{kg}$, $r \approx 6.37 \times 10^{6}\ \mathrm{m}$), $g = GM/r^2 \approx 9.8\ \mathrm{m/s^2}$. The apple accelerates downward because it is driven by coupling between its own refresh phase and the Earth’s λ gradient. It falls not because of “force” or “curvature”, but because of a **push**: the faster refresh (lower λ) on the side away from Earth pushes the apple toward the slower-refresh (higher λ) vacancy on the near side.

Part III: Essence of Gravity – Macroscopic Projection of Refresh Gradient

From the above derivation, we redefine the essence of gravity.

Gravity is neither a force nor curvature, but the macroscopic projection of the gradient of refresh delay.

- Essence of force: In TCM, force is redefined as $\boldsymbol{F} = -mc^2 \nabla(\ln \lambda)$. It is no longer Newton’s “action at a distance” nor Einstein’s “geometric curvature”, but the natural drift tendency of an object (high-λ region) in a λ-field gradient.
- Emergence of Newton’s law: As shown, under weak-field, low-speed limits, TCM’s gravity formula rigorously reduces to Newton’s gravitation $F = GMm/r^2$.
- Equivalence to Einstein’s “curvature”: Although TCM space is absolutely rigid, the λ-field gradient deflects paths of photons and matter. This deflection is mathematically equivalent to the geodesic equation in general relativity. All classic tests of GR (perihelion precession of Mercury, light deflection) have corresponding λ-gradient-based explanations in TCM, with different physical ontology.

Part IV: Unified Picture of Gravity – From Apple to Galaxy

TCM-NAN(λ) unifies all gravitational phenomena from microscopic to cosmic scales within one mechanism.

- Local (apple, GPS): Acceleration directly given by $a = -c^2 \nabla(\ln \lambda)$. The GPS calibration formula

$$\Delta t_{\mathrm{grav}} = T_0 \cdot \frac{GM}{c^2} \left(\frac{1}{r_{\mathrm{earth}}} - \frac{1}{r_{\mathrm{GPS}}} \right) \times 86400$$

is the exact integral result, perfectly reproducing time dilation from λ gradient, with no curved spacetime assumption.

- Galactic scale (dark matter replacement): At galaxy edges, visible matter is dilute, and $\nabla\lambda$ is not fully determined by visible mass $M_{\text{vis}}(r)$, but also includes energy from cosmic initial velocity difference Δv_0 "hidden" in λ gradient during long-term evolution. This produces an extra constant centripetal acceleration explaining observed flat rotation curves:

$$v^2(r) = \frac{GM_{\text{vis}}(r)}{r} + \alpha T_0 \frac{\kappa}{r} \tag{7}$$

Precise fitting of $v_{\text{flat}} \approx 220\ \text{km/s}$ means WIMP dark matter particles are no longer needed; they are merely projections of λ gradient at galactic scales.

- Cosmic scale (dark energy replacement): On cosmic scales, steady-state creation effects of λ gradient (described by source term $Q = \alpha \cdot (\nabla\lambda) \cdot \rho_m$) provide effective negative pressure driving accelerated expansion. This completely replaces the mysterious dark energy constant Λ, explains the origin of $H_0 \approx 70.9$, and naturally resolves tensions in Hubble constant H_0 and matter clustering parameter S_8.

Part V: Conclusion – Panoramic Solution of Gravity

Within TCM-NAN(λ), we obtain an exhaustive and self-consistent solution to gravity:

1. **Origin of gravity**: The root of gravity is the spatial inhomogeneity ($\nabla\lambda$) of the refresh delay field λ. λ itself is excited by matter (mass) and measures refresh frequency of spacetime grids.
2. **Mechanism of gravity**: An object (e.g., apple) in the λ field spontaneously drifts toward regions of larger λ (slower refresh) in each reappearance to preserve wavefront continuity. Macroscopically, it appears pushed toward the mass center. Equation of motion:

$$\boldsymbol{a} = -c^2 \nabla(\ln \lambda) \tag{8}$$

3. **Unity of gravity**: From apple falling (Newton) to GPS calibration (time dilation), galactic rotation curves (dark matter replacement) to cosmic acceleration (dark energy replacement), all gravitational phenomena at all scales are expressions of the same λ field and its differential forms in different environments. 95% of the universe's "missing" energy (dark matter + dark energy) is essentially projection of velocity-difference hidden energy in λ gradient.
4. **Essence of gravity**: Gravity is neither force nor curvature, but a necessary "phase-correction" motion of the cosmic refresh engine running its fundamental code to compensate for uneven local refresh rates.

Final Conclusion

An apple falls not because the Earth “pulls” it, nor because spacetime “curves” it, but because the whiteboard region around it refreshes faster than near Earth, pushing it toward slower-refresh regions. This simple physical picture is rooted in the logical necessity of NAN (Nothing Allows Nothing), driven by a tiny initial difference ε, and ultimately forms the entire gravitational universe we perceive – from apples to galaxies.

Comparison of Three Theories of Apple Falling

Dimension	Newtonian Mechanics	General Relativity	λ-dynamics
Core idea	Apple falls due to universal gravity, action at a distance, inverse-square force	Gravity is not force; mass curves spacetime, apple follows geodesic	Gravity is refresh pressure difference from λ gradient, phase correction on rigid whiteboard
Formula	$F = G\frac{Mm}{r^2}$, $a = \frac{GM}{r^2}$	Einstein field equation $G_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}$	$\boldsymbol{a} = -c^2\nabla(\ln\lambda)$, $\lambda(r) = 1 + \frac{GM}{c^2 r}$
Spacetime	Absolute, uniform, fixed background	Dynamical continuum, mass-energy curves spacetime	Rigid discrete whiteboard, no expansion or curvature
Cause of fall	Attractive force from Earth pulls apple downward	Earth curves spacetime; geodesic motion appears as falling	Faster refresh on far side pushes apple toward slower-refresh near side
Physical definition	Action at a distance without medium	Geometric effect of spacetime curvature	Phase correction of cosmic refresh engine from refresh pressure difference
Limitation	Weak-field, low-speed only	Conceptually inconsistent with force mechanism	Unified across all scales, no dark components needed

From Newton’s “pull”, to Einstein’s “curve”, to λ-dynamics’ “push” from refresh pressure difference, humanity’s understanding of apple falling advances from appearance to essence, from continuous spacetime to discrete quantum information. This is not only progress in physics, but also repeated reconstruction of the human worldview.